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whole sample, they would tend to obscure the delay-time aspect of the response.

Some mention should be made of the possibilities for the simultaneous determination of the heat capacity, in order to permit calculation of the thermal conductivity. A number of thermal diffusivity methods in use have been elaborated by added measurements which essentially permit calculation of the amount of heat absorbed to cause a given temperature change. These often involve the use of electrical heating of the sample, in which case the heat input is readily measured, and one needs only to make the temperature measurements numerically accurate.[8,13]

In the case of the heat-wave methods, one should be able to gain similar information from the slope of the rising part of the curve, provided that the emissivity of the radiating surfaces is known. A similar approach was used by Parker *et al.*[1] They coated the sample with a layer of carbon, which is expected to provide the surface with known and reproducible emissivity. As noted in Sec. IV, the slope depends primarily on the value of hV/Cw so that a knowledge of w, the temperature and emissivity (which give h) allow one to determine C. Parker *et al.* used the over-all temperature rise, which is practical for their experiment since a limited amount of heat is supplied.

In summary, the heat-wave method can be related generically to many other methods for measuring thermal diffusivity. Its particular advantage is the possible simplicity in application. The various elements of the method, however, bear relation to other methods.

[13] D. C. Wallace, P. H. Sidles, and G. C. Danielson, J. Appl. Phys. **31**, 168 (1960).

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Temperature and Finite Pulse-Time Effects in the Flash Method for Measuring Thermal Diffusivity*

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The flash technique for measuring thermal diffusivity is analyzed for the case of a cylindrical-shaped specimen of radius r_0 and thickness a to determine the effects of radiation at high temperatures, finite duration of the heat pulse, and the feasibility of low temperature measurements. It is found that the flash diffusivity method is useful in two complementary limits: (1) pulse time τ short compared to the characteristic thermal response time t_c, (2) τ/t_c of the order 1 to 10. The former case corresponds to the original description of Parker, Jenkins, and Abbott, while the latter case is suitable at very low temperatures. Moreover, it is shown that there is an optimum specimen thickness a for a given material and pulse time τ, in the sense that a higher temperature can be reached before any corrections have to be made to the Parker *et al.* analysis.

I. INTRODUCTION

RECENTLY, a new technique has been proposed for the measurement of thermal diffusivity at high temperatures.[1] Small transient departures from ambient temperature are produced by a pulse of energy (viz., optical flash tube, laser pulse, or electron beam) from a high-intensity source. The pulse of energy is absorbed on one face of the sample which is in the form of a thin wafer. The subsequent transient response of the opposite face is then related to the thermal diffusivity of the sample. Parker *et al.*[1] gave the particularly simple relation

$$\alpha_{\frac{1}{2}}=1.37a^2/\pi^2t_{\frac{1}{2}}, \tag{1}$$

where $t_{\frac{1}{2}}$ is the time required for the back face of the sample to reach half-maximum in its small temperature excursion and a is the sample thickness. The derivation of Eq. (1) rests on approximations which may be quite restrictive. In particular, the possibility of radiation losses from the sample was not considered. Moreover, the radiation pulse from the flash lamp was assumed to be of negligible duration in comparison with the characteristic rise time for the temperature excursion of the sample.

In the following, these considerations are analyzed and nearly exact equations are given together with criteria for the elementary description corresponding to Eq. (1).

* This work was supported in part by the Advanced Research Projects Agency, U. S. Department of Defense.

[1] W. J. Parker, R. J. Jenkins, C. P. Butler, and G. L. Abbott, J. Appl. Phys. **32**, 1679 (1961).

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II. GENERAL ANALYSIS

We consider a sample in the form of a cylinder which is illuminated with a pulse of radiation incident on the front face of the sample. We denote the front and back of the sample as $x=0$ and $x=a$, respectively, and the radius of the sample by r_0.

The heat flow in the sample is governed by[2]

$$-\nabla^2 T(x,r,t)+\alpha^{-1}\dot{T}(x,r,t)=k^{-1}q(x,r,t), \tag{2}$$

where T is the temperature, q is the heat-source term, α is the thermal diffusivity, k is the thermal conductivity, and t is the time. ∇^2 is the Laplacian operator in cylindrical coordinates and the dot denotes time derivative. We expect q to have the form $q_0U(x)V(r)W(t)$, where $U(x)$, $V(r)$, and $W(t)$ denote the spatial and temporal distribution of the absorbed incident energy. The constant q_0 has the dimensions of energy/area and represents the amplitude of the heat pulse. Initially, the temperature distribution is considered to be steady-state with no large gradients so that k may be taken to be constant. At time $t=0$, a pulse of radiation is absorbed at $x=0$, so that the temperature of the face at $x=a$ rises slightly above ambient. The temperature behavior is given by the solution of Eq. (2) with appropriate boundary conditions. For convenience, we put $\delta=T-T_0$ and Eq. (2) becomes

$$-\nabla^2\delta(x,r,t)+\alpha^{-1}\dot{\delta}(x,r,t)=k^{-1}q(x,r,t). \tag{3}$$

Under usual experimental conditions, the sample rises in temperature only slightly above its ambient temperature T_0 and the boundary conditions for $\delta(x,r,t)$ derived from the Stefan–Boltzmann radiation law can be written as

$$\delta_x'(0,r,t)=\nu_x\delta(0,r,t), \tag{4a}$$

$$\delta_x'(a,r,t)=-\nu_x\delta(a,r,t), \tag{4b}$$

and

$$\delta_r'(x,r_0,t)=-\nu_r\delta(x,r_0,t), \tag{5}$$

where

$$\nu_{x,r}=4\sigma\epsilon_{x,r}T_0^3k^{-1} \tag{6}$$

and we have used the notation $d\delta/dx=\delta_x'$, $d\delta/dr=\delta_r'$. In Eq. (6), σ and ϵ denote the Stefan–Boltzmann constant and total emissivity of the sample, respectively.

A method for solving Eq. (3) subject to the boundary conditions (4), (5) is given in Sneddon's book on Fourier transforms.[3] The temperature at the back face of the sample, $x=a$, is found to be

$$(k/q_0)\delta(a,r,t)=(2/r_0^2)\sum_{i=0}^{\infty}[z_i^2/(Y_r^2+z_i^2)]\times[J_0(z_ir/r_0)/J_0^2(z_i)]\bar{V}(z_i)\bar{F}(z_i,t). \tag{7}$$

The quantities appearing in Eq. (7) are defined as follows:

$$Y_r\equiv\nu_r r_0=4\sigma\epsilon_r T_0^3k^{-1}r_0, \tag{8}$$

the z_i are the roots of $Y_rJ_0(z_i)=z_iJ_1(z_i)$, (9)

and

$$\bar{V}(z_i)=\int_0^{r_0}drrJ_0(z_ir/r_0)V(r) \tag{10}$$

is the finite Hankel transform of $V(r)$, the radial part of the heat-source function $q(x,r,t)$.

$$\bar{F}(z_i,t)=\sum_{m=0}^{\infty}\int_0^a dyU(y)\int_0^t dt'W(t')R_m(y,z_i,t-t'), \tag{11}$$

where

$$R_m(y,z_i,t)=C_m[X_m\cos(X_my/a)+Y_x\sin(X_my/a)]e^{\omega_{im}t/t_c}, \tag{12}$$

with

$$\omega_{im}=-(a/\pi)^2(X_m^2/a^2+z_i^2/r_0^2) \tag{13}$$

and

$$C_m=(2\alpha/a)(-1)^mX_m(X_m^2+2Y_x+Y_x^2)^{-1}, \tag{14}$$

and the X_m are the roots of

$$(X_m^2-Y_x^2)\tan X_m=2X_mY_x, \tag{15}$$

where

$$Y_x\equiv\nu_x a=4\sigma\epsilon_xT_0^3k^{-1}a. \tag{16}$$

In Eq. (12), the characteristic time t_c is $(a/\pi)^2\alpha^{-1}$, which is roughly the amount of time necessary for a heat pulse to propagate the length of the specimen. A similar expression using $R_m(a-y,z_i,t)$ in (11) gives the temperature at the front face, $\delta(0,r,t)$.

We now examine the case where the incident radiation is absorbed in a very thin layer at the surface of the specimen,[4] and take the incident flux to be uniform over the surface of the specimen.[5] One therefore has

$$U(y)=\delta(y)^6 \quad \text{and} \quad V(r)=1. \tag{17}$$

In this case, Eq. (7) reduces to

$$(k/q_0)\delta(a,r,t)=\sum_{m=0}^{\infty}C_mX_m\sum_{i=0}^{\infty}D_i(r,Y_r)\int_0^t dt'W(t')e^{\omega_{im}(t-t')/t_c}, \tag{18}$$

where we have put

$$D_i(r,Y_r)=[2Y_r/(Y_r^2+z_i^2)]J_0(z_ir/r_0)/J_0(z_i). \tag{19}$$

[2] Equation (2) is valid when the thermal diffusivity is independent of temperature. In the present case, the approximation is justified if the maximum temperature excursion in the sample is kept small. This is discussed in Ref. 1.

[3] I. N. Sneddon, *Fourier Transforms* (McGraw-Hill Book Company, Inc., New York, 1951), pp. 83–84, 176, 177.

[4] For example, $\beta a\gg1$, where $\beta=\int\beta_\lambda U_\lambda d\lambda/\int U_\lambda d\lambda$ is the average absorption coefficient. Here U_λ is the Planck distribution function.

[5] This may be achieved by the use of a laser with a sample of the same radius as the laser beam. However, if a conventional flash tube is used and precautions are taken to insure that the beam is diffuse over the face of the specimen, one would not expect any significant difference in the flow of the heat pulse through the sample.

[6] The Dirac delta function; it should not be confused with $\delta(x,t)=T-T_0$.

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When $Y_r \leq 1$ it is easy to show that the first term in Eq. (18), $(i=0)$, is by far the dominant one for all a/r_0. One may therefore write

$$(k/q_0)\delta(a,r,t) = D_0(r,Y_r)\sum_{m=0}^{\infty} C_m X_m \int_0^t dt' W(t') e^{\omega_{0m}(t-t')/t_c}. \quad (20)$$

In the special case where the duration of the heat pulse is short compared to the characteristic time $t_c = (a/\pi)^2\alpha^{-1}$, $W(t') \cong \delta(t')$,[6] and Eq. (20) becomes

$$(k/q_0)\delta(a,r,t) = D_0(r,Y_r)\sum_{m=0}^{\infty} C_m X_m e^{\omega_{0m}t/t_c}. \quad (21)$$

For $Y_r = 1$, the largest neglected terms in Eq. (18) in the sum on i constitute approximately 4% error in the total summation at $t=t_c$ and correspondingly less for longer times, and for smaller Y_r. Since the measurement of α is determined by observations at $t > t_c$, it is clear that for $Y_r < 1$ Eqs. (20) and (21) are sufficiently accurate.

Low-Temperature Limit Y_x, $Y_r \ll 1$

At low temperatures where Y_x, $Y_r \ll 1$, Eq. (21) reduces to[7]

$$(k/q_0)\delta(a,r,t) = (\alpha/a)[1+2\sum_{m=1}^{\infty}(-1)^m e^{-m^2 t/t_c}]. \quad (22)$$

This is the result obtained by Parker *et al.*[1] A plot of the function $V \equiv \delta(a,r,t)/\delta_{\max}$ is shown as curve (1), Fig. 1. The temperature rises from ambient T_0 to a constant final temperature $T_0 + \delta_{\max}$.

High-Temperature Limit with Radial Shielding

When Y_x, $Y_r \lesssim 1$, Eq. (9) and Eq. (15) may be solved to give

$$\begin{aligned} z_0 &\approx (2Y_r)^{\frac{1}{2}}(1 - Y_r/8 + \cdots), \\ X_0 &\approx (2Y_x)^{\frac{1}{2}}(1 - Y_x/12 + 7Y_x^2/288 + \cdots), \\ X_m &\approx m\pi + 2Y_x/m\pi - 4Y_x^2/(m\pi)^2 - 2Y_x^3/3(m\pi)^3 \\ &\quad + 16Y_x^4/(m\pi)^5 + \cdots m = 1, 2, \cdots. \end{aligned} \quad (23)$$

Equation (21) can now be written explicitly [to first order in the expansions (23)], as

$$k/q_0\delta(Y_x,Y_r,t) = D_0(r,Y_r)[C_0X_0e^{-2(Y/\pi^2)t/t_c} + e^{2(Y_r/\pi^2)(a/r_0)^2 t/t_c}\sum_{m=1}^{\infty} C_m X_m e^{-(m^2+4Y/\pi^2)t/t_c}], \quad (24)$$

where we have put $Y \equiv Y_x + (a/r_0)^2 Y_r$. For the radially adiabatic case $[(a/r_0)^2 Y_r \ll Y_x)]$, numerical values of the function $V \equiv \delta(Y_x,0,t)/\delta_{\max}$ have been calculated with the aid of Eqs. (23) and are shown in Fig. 1. In the case $Y=0$, one sees from Fig. 1 that $\delta(Y,0,t)/\delta_{\max} = \frac{1}{2}$, where $t/t_c = 1.37 = t_{\frac{1}{2}}/t_c$, and since $t_c = a^2/\pi^2\alpha$, the thermal diffusivity is given by Eq. (1). At high temperatures, curves (2), (3), and (4) of Fig. 1 illustrate the effect of radiation. For example, at $Y=1$, $t_{\frac{1}{2}}/t_c = 0.97$. In this case, Eq. (1) would give a value of α in error by roughly 50%.

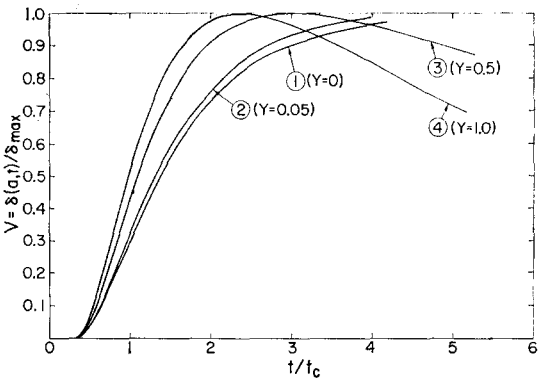


FIG. 1. The temperature rise at the back face of the sample. The curves are normalized to unity at the maximum of the temperature excursion. Note how the value of t/t_c at $V = \frac{1}{2}$ decreases as the radiation parameter Y increases.

One now notes that in Eq. (24), the terms $m \geq 1$ contribute not more than approximately $\frac{1}{2}$ the value of the total sum for all $t \gtrsim t_c$. This means that if $e^{2/\pi^2(a/r_0)^2 Y_r} \lesssim 1.1$, i.e., $(a/r_0)^2 Y_r \lesssim 0.5$, the function $\delta(Y,0,t)$ [where $Y = Y_x + (a/r_0)^2 Y_r$], is within 5% of the value of $\delta(Y_x,Y_r,t)$ for all $t \gtrsim t_c$. Thus, even in the extreme case $Y_x = 1$, providing the use of radial shielding and suitable choice of a/r_0 ensure that $(a/r_0)^2 Y_r \lesssim 0.5$, the curves of Fig. 1 are applicable even when radial heat loss is not negligible.

In practice, the usefulness of the results shown in Fig. 1 is realized by determining an initial value of the diffusivity α_0 from Eq. (1), then estimating Y from a reasonable estimate of the emissivities ϵ_x, ϵ_y, and the heat capacity ($k = \rho c \alpha_0$, where ρ is the density and c the heat capacity). One then re-evaluates α by use of the curves in Fig. 1. This iterative procedure may be repeated to obtain a final value for α.

Figure 2 shows how the numerical factor $t_{\frac{1}{2}}/t_c$ varies from 1.37 as Y takes on successively greater values. One notes that for Y less than 5×10^{-2}, the approximation, Eq. (22) and hence Eq. (1), is accurate to about 5% or better. By way of illustration, Table I lists some temperatures corresponding to this value of Y for different values of $k/\epsilon a$ where we have put $\epsilon = \epsilon_x + (a/r_0)\epsilon_r$ for convenience. For typically high conductivity materials such as metals, $k/\epsilon a$ is easily made greater than 1.0 cal/cm² sec °K. In this case radiation effects may be negligible for all temperatures below the melting point. The most restrictive case cited, $k/\epsilon a = 0.1$, is representative of typically poor thermal conductors (e.g., the alkaline-earth oxides), with k

[7] Note that as $Y_x \to 0$, the correct limits derived from (14) and (23) are: $C_m X_m \to 2(\alpha/a)(-1)^m$, $C_0X_0 \to \alpha/a$, $C_m Y_x \to 0$, and $C_0 Y_x \to 0$.

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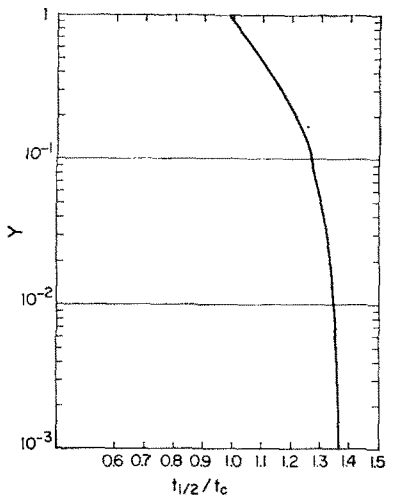


FIG. 2. This illustrates how the numerical factor of Eq. (1) (t/t_c at $V=\frac{1}{2}$) decreases as the radiation parameter $Y=Y_x+a^2Y_r/r_0^2$ increases. With a reasonable estimate of Y, the value of $t_{\frac{1}{2}}/t_c$ taken from this curve may be used in Eq. (1).

$\approx 10^{-2}$ cal/cm² °K sec and $a=10^{-1}$ cm. However, it is often possible to make the sample an order of magnitude thinner, viz., 10^{-2} cm so that $k/\epsilon a\approx 1$. However, in the following paragraphs, it is shown that the use of thin samples makes necessary a correction arising from a finite duration of the heat pulse.

III. FINITE PULSE-TIME EFFECTS

In some applications of the flash heating technique, the duration of the energy pulse is comparable to the characteristic thermal diffusion time t_c in which case, the rise in temperature of the specimen is expected to be retarded.

To estimate the "finite pulse-time effect," it is sufficient to treat the pulse as a simple square wave of width τ. One therefore assumes

$$q(y,r,t)=q_0\delta(y)\tau^{-1}, \quad t\leq\tau$$
$$=0, \quad t>\tau. \tag{25}$$

For $t>\tau$, Eq. (20) becomes

$$(k/q_0)\delta(a,r,t)$$
$$=D_0(r,Y_r)\sum_{m=0}^{\infty}C_mX_me^{\omega_{0m}t/t_c}\left[\frac{1-e^{-\omega_{0m}\tau/t_c}}{\omega_{0m}\tau/t_c}\right]. \tag{26}$$

From Eq. (26) it is clear that when the "width" of the heat pulse $\tau\ll t_c$ the finite pulse-time effects are negligible since Eq. (26) reduces to (21).

TABLE I. Maximum temperatures below which Eq. (1) is accurate to less than 5% ($Y<5\times10^{-2}$).

$k/\epsilon a$ (cal/cm² °K sec)	$T_{0\max}$(°K)
5.0	3680
2.0	2710
1.0	2150
0.5	1710
0.2	1260
0.1	1000

The results of a more realistic calculation for the case where the heat pulse is produced by the radiation from a flash lamp are shown in Fig. 3. The heat pulse from such a device can be adequately approximated by the function[1]

$$q(t)=q_0[(\tau-t)/\tau^2], \quad 0\leq t\leq\tau$$
$$=0, \quad t>\tau. \tag{27}$$

Here the pulse time τ, which is of the order of 10^{-3} sec, is characteristic of a given lamp and may be measured in the laboratory. For the function (27), Eq. (20) was programmed for an IBM 7090 computer and some of the results for the case $Y=0$ are shown in Fig. 3. One sees that for finite pulses, the correction to Eq. (1) is in the opposite sense to that caused by radiation losses. That is, as the pulse time τ becomes comparable to t_c, the numerical factor appearing in Eq. (1) *increases*. This is shown explicitly for a pulse of the type (27) in Fig. 4.

IV. APPLICATION AT LOW TEMPERATURES

Near the absolute zero of temperature the specific heat of all substances tends to zero and one generally finds the drop off in the specific heat exceeds that of the thermal conductivity so that the thermal diffusivity tends to much larger values. Consequently, while radiation effects are not important at low temperatures in the application of the flash diffusivity technique, the finite pulse-time effect may be prohibitively restrictive. For some substances, the diffusivity may become as high as 10^4 cm²/sec so that t_c may be as small as 10^{-5} sec for a 1-cm-thick specimen. The criterion $\tau/t_c\ll 1$ cannot, in this circumstance, be obeyed outside of the difficult expedient of providing extremely short (viz., microsecond) heat pulses. An alternate procedure employing square-wave pulses

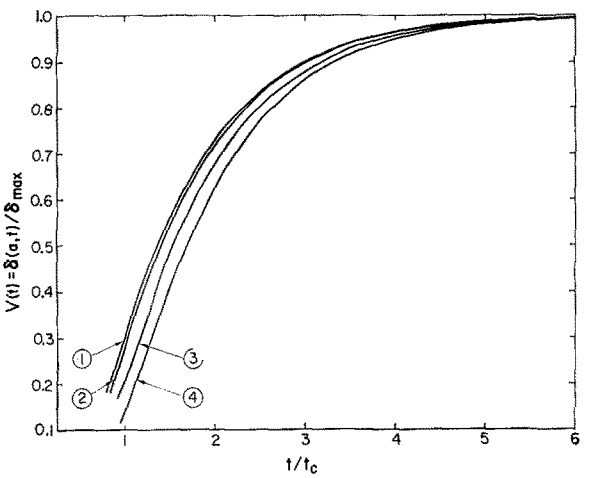


FIG. 3. The temperature rise at the back face of the sample normalized to unity at the temperature maximum. These curves illustrate the finite pulse-time effect for a pulse of the form of Eq. (27). Curve (1) is for $\tau=0.01t_c$, curve (2), $\tau=0.10t_c$, curve (3), $\tau=0.51t_c$, and curve (4), $\tau=0.91t_c$.

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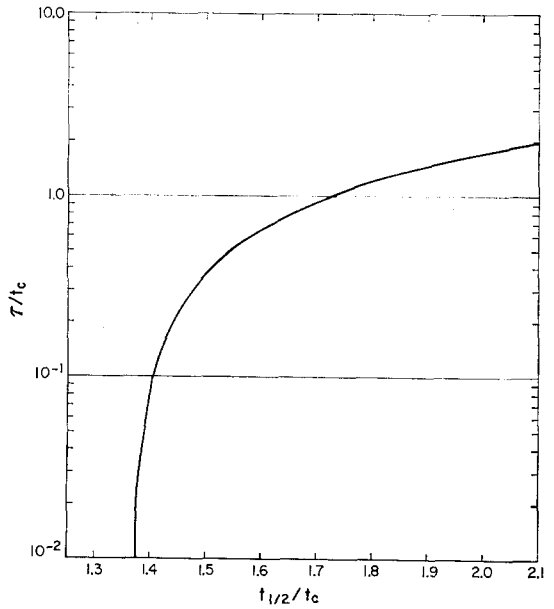


FIG. 4. This shows how the numerical factor $t_{\frac{1}{2}}/t_c$ of Eq. (1) increases as the pulse time increases. The values are for a pulse of the form of Eq. (27). Note that as long as $\tau/t_c<10^{-1}$, the finite pulse-time effect is negligible.

longer than the characteristic time t_c, may be realized as follows. In the nonradiative case, it is easily shown that for $\tau/t_c\gg1$, Eq. (26) gives

$$k\delta(\tau)/q_0\simeq\alpha/a(1-\pi^2t_c/6\tau). \tag{28}$$

Equation (28) is accurate to approximately 2% at $\tau/t_c=4$ and more accurate for larger τ/t_c.

Noting that $k\delta(\infty)/q_0=\alpha/a$, one has

$$1-\delta(\tau)/\delta(\infty)\simeq a^2/6\alpha\tau. \tag{29}$$

The quantity on the right-hand side is of conveniently measurable magnitude (say approximately equal to $\frac{1}{2}$) for conveniently sized samples when $\alpha\tau\approx1$ so that thermal diffusivities as large as 10^2 cm^2/sec may be measured using heat pulses as long as 10^{-2} sec.

This procedure may be particularly useful for the study of alloys where the thermal diffusivity is often substantially smaller than the diffusivities of the parent metals. In some instances, t_c may be sufficiently long that electrical heating may be used. For example, when $\alpha\leq1$ cm^2 sec, the pulse time τ may be 1 sec or longer. This should be easily accomplished with a small electrical heater. Using this technique, an apparatus of the type described by Morin and Maita[8] ought to permit simultaneous measurement of the heat capacity C and the thermal diffusivity α.

V. DISCUSSION

The flash diffusivity method is useful in two complimentary limits: (1) in the limit introduced by Parker *et al.*[1] when $\tau/t_c\ll1$, and (2), for square-wave pulses, when τ/t_c is of order 1 to 10. The former procedure is advantageous at high temperatures. With the formulas derived in the foregoing text, it is possible to make suitable corrections for both the effects of radiation and the finite pulse-time effect. Moreover, it is possible to optimize the sample thickness so as to minimize corrections to Eq. (1). Whereas the radiation effects are made smaller by *decreasing* the sample thickness, the ratio τ/t_c (and hence the finite pulse-time effect), is made smaller by *increasing* the sample thickness. Consequently, for a given material, and for a given pulse time τ, there will be an optimum sample thickness in the sense that a higher temperature can be reached before any correction to Eq. (1) is required. For example, suppose the pulse width is 0.001 sec. For a typical metal, we may have $k/\epsilon a\approx0.5/a$, and $\tau/t_c\approx0.004/a^2$. In this case, an optimum thickness is approximately 0.2 cm. Using Table I, we note that this corresponds to 5% accuracy at slightly above 3000°K.

On the other hand, for a typical oxide, we may have $k/\epsilon a\approx0.005/a$ and if $\tau\approx0.001$ sec, $\tau/t_c\approx0.0001/a^2$. By comparison with Fig. 4, one sees that to avoid finite pulse-time effects for this case, a thickness of at least 0.015 cm is required. With this thickness, it is clear that Eq. (1) is in error by about 8% at 2000°K [Cf., Eq. (8) and Fig. (2)]. However, if an energy pulse of duration much less than 10^{-3} sec is employed, the use of thinner samples is permitted and Eq. (1) will be accurate to correspondingly higher temperatures.

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[8] F. J. Morin and J. P. Maita, Phys. Rev. **129**, 1115 (1963).

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